

ARUNACHALAM K

Robotics Engineer | Embedded Systems | AMR / AGV Development | Open to Coimbatore
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SUMMARY

Final-year B.E. Robotics & Automation Engineering (Honours) student (2023–2027, CGPA 7.6) with strong hands-on experience in autonomous robot development, sensor integration (LiDAR, IMU, Ultrasonic), ROS2, SLAM, and embedded systems. Built and led a quadruped surveillance robot with real-time computer vision and autonomous navigation. Proficient in PID control, motor driver circuits, PCB design (KiCad), soldering, and hardware prototyping. Founder of TruRock robotics startup. Passionate about AMR/AGV systems and intelligent automation. Open to immediate joining and relocation to Coimbatore.

TECHNICAL SKILLS

Robotics & Navigation: ROS2, SLAM, Nav2 Stack, autonomous navigation, sensor fusion, path planning, Gazebo simulation, RViz, AMR/AGV systems, behaviour trees
Sensors & Hardware: LiDAR, IMU, Ultrasonic, IR, Magnetic sensors, encoders, Servo/DC motors, motor drivers (L298N, L293D), ESCs, actuators, electric motors, LiPo power systems
Embedded & Control: Arduino, Raspberry Pi, Embedded C, PID tuning, motor control, PWM, circuit assembly, soldering, terminal crimping, wiring, sensor interfacing
Programming: Python, C++, Embedded C, HTML, CSS, JS | Libraries: OpenCV, NumPy, YOLO (computer vision)
PCB & CAD Design: KiCad (PCB design & schematic capture), SolidWorks (3D CAD), TinkerCAD, Bambu Lab Studio (3D Printing)
Tools & OS: Linux (Ubuntu), Windows, Git, VS Code, MS Office, Gazebo, RViz

EXPERIENCE

Robotics & IoT Project Developer -- SIVACK Technologies *Apr 2025 – Present*

- Developed IoT and robotics automation projects involving sensor integration, embedded control systems, motor drivers, and circuit assembly for industrial applications.
- Assembled and tested robotics hardware including Arduino-based control boards, sensor arrays, and actuator wiring; performed hands-on soldering and component integration.
- Delivered technical training sessions on embedded systems and robotics for students; developed hands-on project demos from scratch.

Founder & CEO -- TruRock, Erode *Jan 2024 – Present*

- Founded robotics R&D startup; led autonomous robot prototyping, embedded system design, PCB schematic work, and hardware assembly - mirrors AMR/AGV product development lifecycle.
- Managed component sourcing, circuit wiring, mechanical assembly, testing, and iteration across multiple robotics projects simultaneously.

PROJECTS

Quadruped Surveillance Robot - Team Leader *Jun 2025 – Present*

- Built 4-legged autonomous robot for hazardous environment monitoring - integrated LiDAR, IMU, and OpenCV/YOLO computer vision for real-time obstacle detection and autonomous navigation.
- Implemented ROS2 navigation with sensor fusion and SLAM; handled full hardware assembly including motor wiring, sensor mounting, power distribution, and embedded controller integration.

Line Follower Robot - Team Leader & Inventor *Jun 2023 – May 2024*

- Designed PID-controlled autonomous line follower with IR sensor array and L298N motor driver; tuned PID gains (Kp, Ki, Kd) for stable, accurate path tracking at competition speed.

OTTO Robot – Assembled & Trained *Jan 2024 – Present*

- Built servo-driven biped with ultrasonic obstacle avoidance and embedded motion control logic; configurable gait parameters — directly applicable to AMR locomotion and motion control systems.

EDUCATION

Degree	Institution	Year	Score
B.E. Robotics & Automation Engineering (Honours)	Erode Sengunthar Engineering College	2023–2027	CGPA 7.6
Class XII	SSV Matric Hr. Sec. School, Sivagiri	2023	80%
Class X	SSV Matric Hr. Sec. School, Sivagiri	2021	80%

CERTIFICATIONS & ADDITIONAL

Certifications: Python – W3Schools | Python Game Development – GUVI | HTML, CSS, JS – W3Schools
Languages: Tamil (Native), English (Professional) |
Interests: AMR/AGV Research, Drone Dev, 3D Printing, Chess